Smart Traffic Management System

A multi-agent AI system: Max-Pressure signals, A* routing, ambulance preemption, negotiated bus priority, hill-climbing tuning

M1 **P.M. Radha Krishna (CB.SC.U4CSE23134)** Environment & Simulation

M2 **Member 2** Intersection Agent

M3 **Member 3** Vehicle Agent & Search

M4 **Member 4** Priority Vehicles

M5 **Member 5** Coordinator & Testing

AI Case Study | Review 1 & Review 2 | Amrita Vishwa Vidyapeetham, Coimbatore

Why fixed-time signals fail

TEAM

Whole team

Three problems, each solved by a different agent

Static timing

The same green goes to an empty side street and to a rush-hour arterial.

Solved by: Intersection agents (Max-Pressure)

Emergency delay

Ambulances lose time at every red light on their way.

Solved by: Ambulance preemption protocol

Information lag

Drivers find out about an accident only when they reach the queue.

Solved by: Coordinator broadcasts + A* re-planning

-47.5%

average wait vs fixed-time (S1)

-29.6%

ambulance travel time, 0 s at red (S3)

96 s

vs 145 s travel past an accident (S4)

Five agents, five owners

Each member owns an agent, its algorithm, its scenario and its tests

TEAM

Whole team

M1	**Environment & simulator**	BPR travel time, saturation flow, N x N grids	S8 scalability, visual demo
M2	**Intersection agent**	Max-Pressure + anti-starvation, green-wave	S1, S2
M3	**Vehicle agent**	A* (vs BFS, UCS, Greedy, W-A*), beliefs	S4, S7, A* trace
M4	**Ambulance + Bus agents**	Preemption; negotiated priority	S3, S5
M5	**Coordinator (TMC)**	Hill climbing, incident broadcast, protocol	S6, 43-test suite

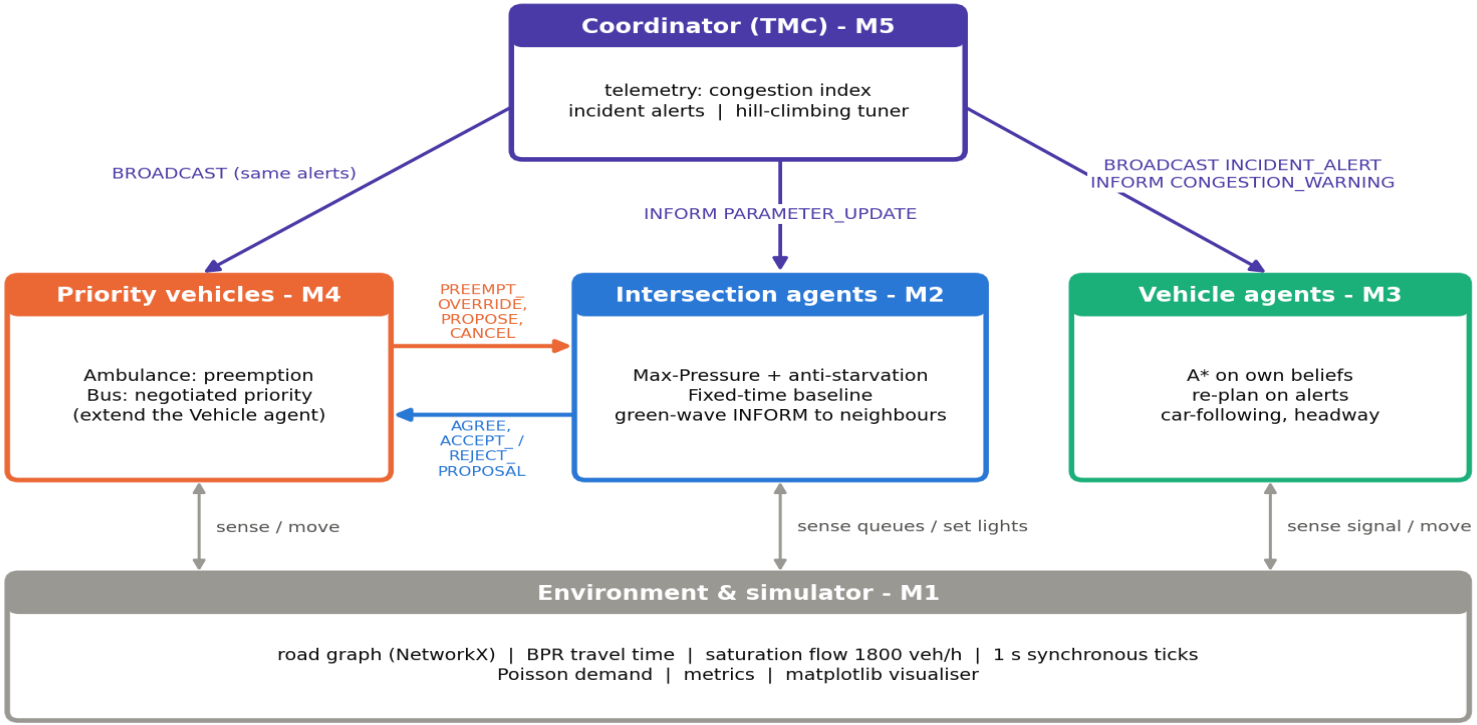
System architecture

TEAM

Whole team

Agents interact only through FIPA-ACL messages via the broker

Every arrow between agents is a FIPA-ACL message through the broker, never a direct function call



PEAS formulation

TEAM

Whole team

Review 1: what each agent optimises, senses and controls

Agent	Performance	Environment	Actuators	Sensors
Intersection	Min delay, max throughput, no starvation, 3 s yellow	4-leg junction, 8 roads, random arrivals	Phase NS / EW / yellow; AGREE, ACCEPT, REJECT	Queue counts on 8 roads; inbox
Vehicle	Min travel time, min stops	Road graph, signals, cars ahead	Accelerate / stop / cross; choose route	Speed, signal, gap; TMC alerts
Ambulance	Min response time, 0 s at red	Roads with background traffic	Drive 60 km/h; PREEMPT, CANCEL	Route look-ahead ETA; AGREE
Bus	Min passenger delay, limited harm	Mixed traffic; junctions may refuse	PROPOSE, CANCEL	Next-junction ETA; replies
Coordinator	Network delay; delay on unseen traffic	Whole network (aggregate)	BROADCAST alerts; PARAMETER_UPDATE	All road densities; incidents

Environment & agent analysis

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Whole team

Review 1: task-environment properties and agent architectures

Partially observable
each agent sees only local roads + messages

Multi-agent
cooperative junctions, competing vehicles

Stochastic
Poisson arrivals, random incidents

Sequential
a green now changes every later queue

Dynamic
queues grow while agents think

Continuous
positions and speeds; time in 1 s ticks

Known physics
roads and BPR known; demand not

Agent	Architecture
Intersection	Utility-based, with internal state
Vehicle	Goal-based, model-based (search)
Ambulance	Goal-based + communication
Bus	Utility-based negotiator
Coordinator	Learning agent (critic = simulation)

Route search: A* over each vehicle's beliefs

M3

Vehicle & Search

Review 1: algorithmic modelling & search strategy

State-space formulation

State: junction the vehicle is at

Actions: drive any road it believes is open

Goal test: s = destination

Step cost: BPR travel time t0 (1 + 0.15 (V/C)^4)

Heuristic: h(n) = straight-line distance / v_max

Evaluation: f(n) = g(n) + h(n)

Admissible: h(n) <= h*(n)

No route is shorter than the straight line, and no road is faster than v_max, so h never over-estimates. Congestion only raises true cost. (UT-01)

Consistent: h(u) <= c(u,v) + h(v)

Triangle inequality plus c(u,v) >= length / v_max. So f never decreases, the first goal popped is optimal, and nothing is re-expanded. (UT-02)

A* step by step

M3

Vehicle & Search

3x3 congested grid, INT_10->INT_11 closed, trip W0 -> E2 (python run_simulation.py --astar-trace)

Step	Expand	g(n) s	h(n) s	f = g + h
1	W0	0.0	64.44	**64.44**
2	INT_00	14.41	51.95	**66.37**
3	INT_01	28.82	40.76	**69.58**
4	INT_10	28.96	45.57	**74.52**
5	INT_02	43.23	32.22	**75.45**
6	INT_11	44.99	32.22	**77.21**
7	INT_12	57.64	20.38	**78.02**
8	INT_22	72.05	14.41	**86.46**
9	E2	86.46	0.0	**86.46**

What to notice

f never decreases: 64.4 -> 86.5 (consistency)

High-h nodes (N0, W1) are never expanded

A*: 9 expansions | UCS: 20 expansions

Same optimal cost: 86.46 s

Route: W0 > INT_00 > INT_01 > INT_02 > INT_12 > INT_22 > E2

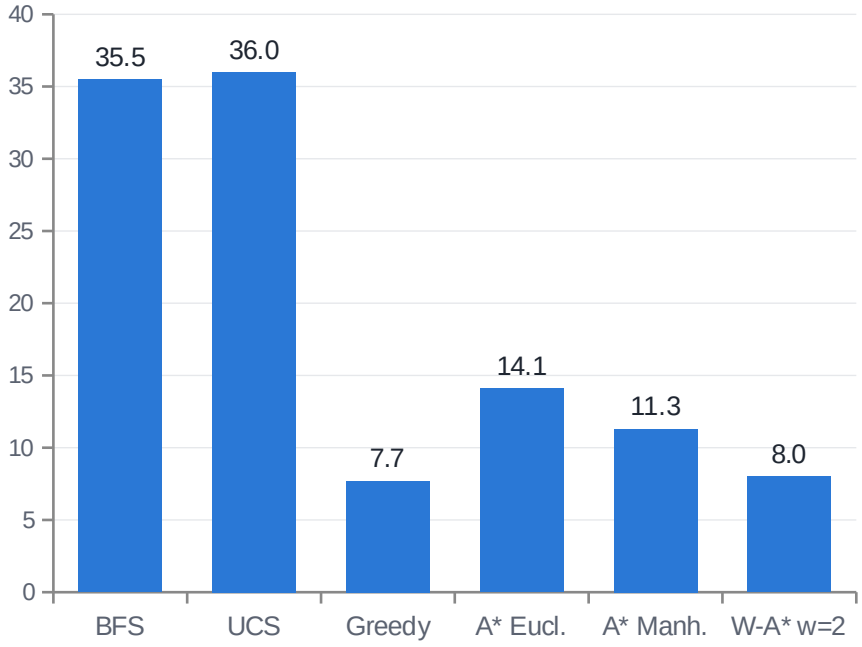
Why A*? Five strategies compared

M3

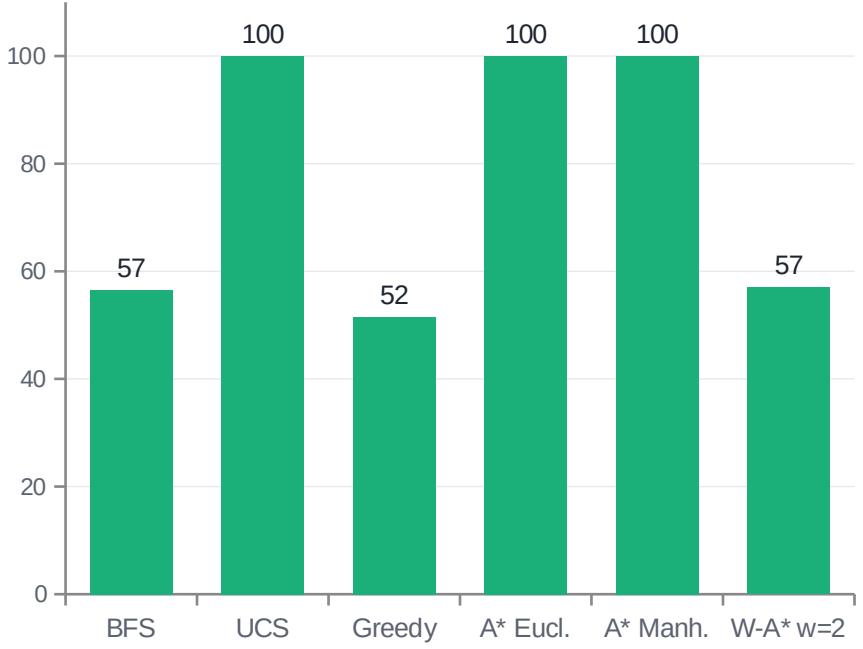
Vehicle & Search

S7: 6x6 (60 nodes, 168 roads), 200 random trips, one closed road

Nodes expanded (lower = faster)



Routes that are optimal (%)



Max-Pressure signal control

M2

Intersection

Review 1: decentralised; each junction uses only its own 8 queues

Algorithm (every second)

Movement pressure: w = Q_in - Q_out

Phase pressure: P = sum of max(0, w)

Anti-starvation: S = P + 0.5 x red-time

After 5 s minimum green:

- red side empty -> keep green
- green >= 30 s -> switch
- S(red) > S(green) + 1 -> switch

Every switch goes through a 3 s YELLOW

INT_00 movement	Q_in	Q_out	max(0, w)
N0 -> south	8	2	6
S -> N0 (north)	3	5	0
W0 -> east	2	1	1
E -> W0 (west)	4	0	4

NS green for 12 s:

S(NS) = 6 + 0.5 x 0 = 6

S(EW) = 5 + 0.5 x 12 = 11

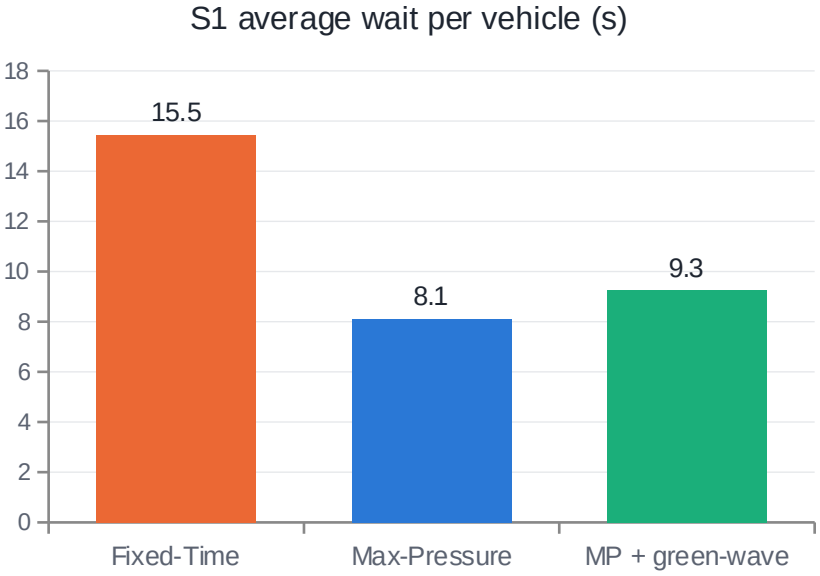
11 > 6 + 1 -> YELLOW, then EW green

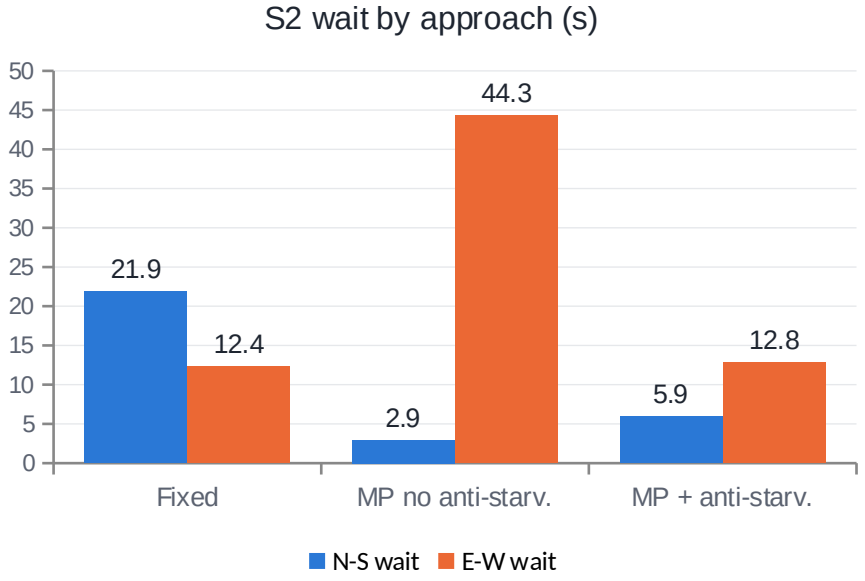
Results: signals

M2

Intersection

S1 2x2 grid, 10 seeds | S2 rush hour, 87% of traffic N-S





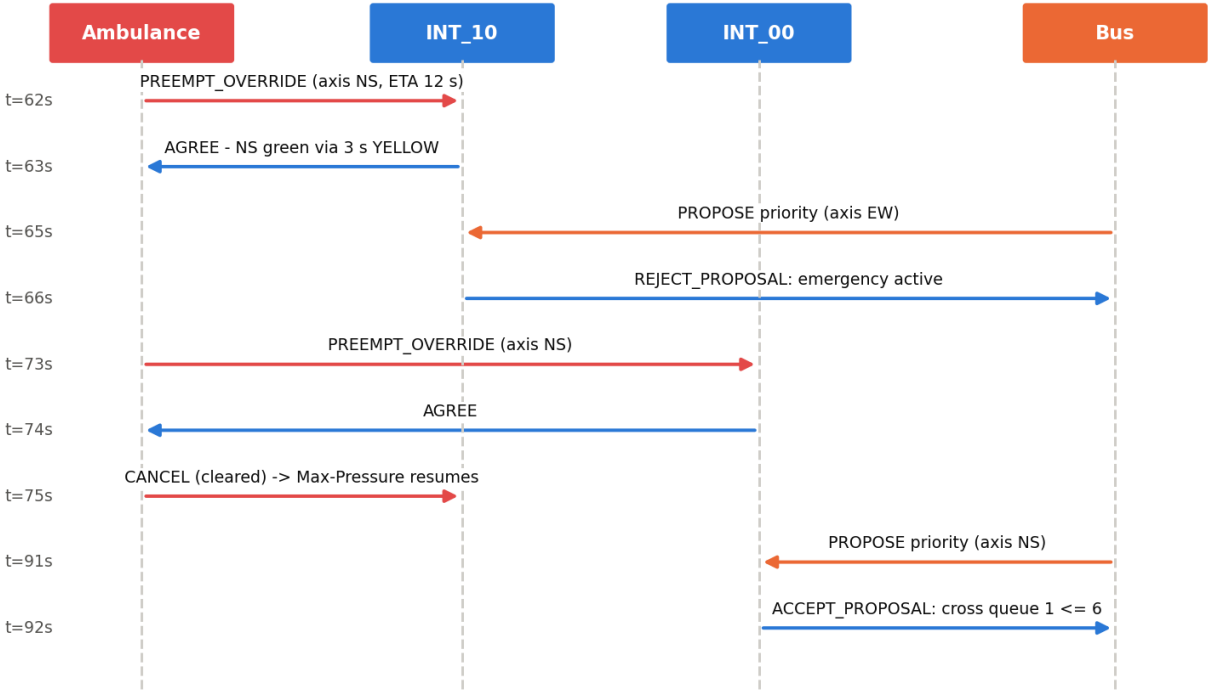
-47.5% wait vs fixed-time. **14 of 108** E-W cars never served without anti-starvation; 0 with it.

Multi-agent interaction, as it happened

M4

Priority vehicles

Review 2: messages recorded in the demo run (python run_simulation.py --protocol)



Two protocols

Ambulance: PREEMPT_OVERRIDE -> AGREE -> CANCEL. Absolute, but always via yellow.

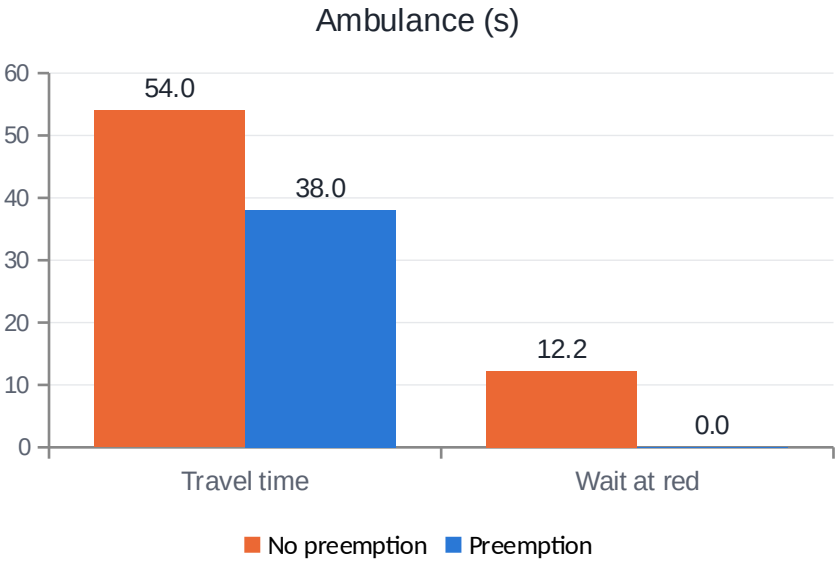
Bus: PROPOSE -> ACCEPT or REJECT. The junction decides; refused here because the ambulance holds INT_10.

Results: priority vehicles

M4

Priority vehicles

S3 same controller, preemption off vs on | S5 bus priority, 4 buses x 40 passengers



Bus priority	Light cross	Heavy cross
Bus travel (s)	63.8 -> 58.0	63.3 -> 59.5
Car wait (s)	2.41 -> 4.76	2.87 -> 4.12
Accepted / rejected	64 / 0	43 / 21
Net passenger-s saved	807	453

Ambulance 29.6% faster, 0 s at red, other cars -0.12 s.

Bus priority: cars wait longer, but more people get through sooner. The junction refuses when the cross street is long.

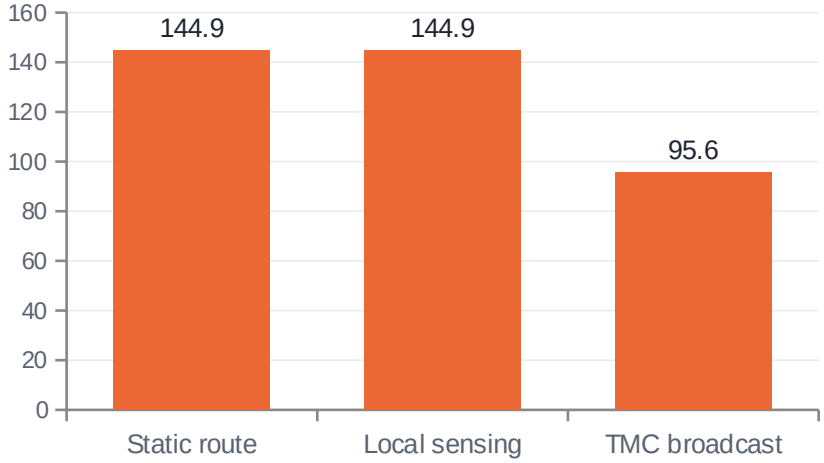
Coordinator: broadcasts and hill climbing

M5

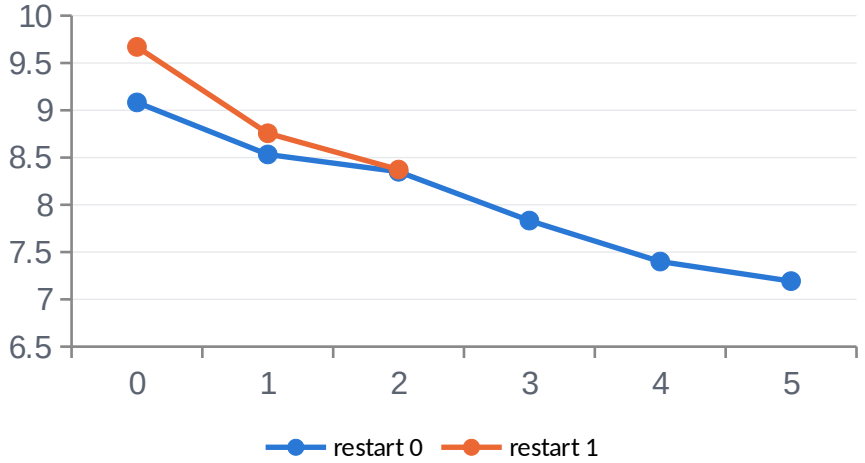
Coordinator

S4 accident on INT_00->INT_01 | S6 tuning signal parameters

Road slowed to 2 m/s: travel time (s)



Hill climbing: avg wait at each step (s)



Closure: static routing strands 12 / 12 cars; A* re-routing delivers 100%. **Tuning:** unseen traffic 8.65 s -> 6.96 s; best of 3 restarts 6.85 s; 62 simulations.

Tools, setup and code structure

M1

Environment

Review 2: tool/package selection & code structure

Tool	Why
Python 3.10+	Readable; standard in AI courses
NetworkX	Graph storage; search written by hand
rich	Readable live console demo
HTML demo page	Route demo, offline
unittest	43 tests, no extra install
Not SUMO / Mesa	Would hide the models we need to explain

```
pip install -r requirements.txt
python run_simulation.py --all
python tests/run_tests.py
python generate_results.py
```

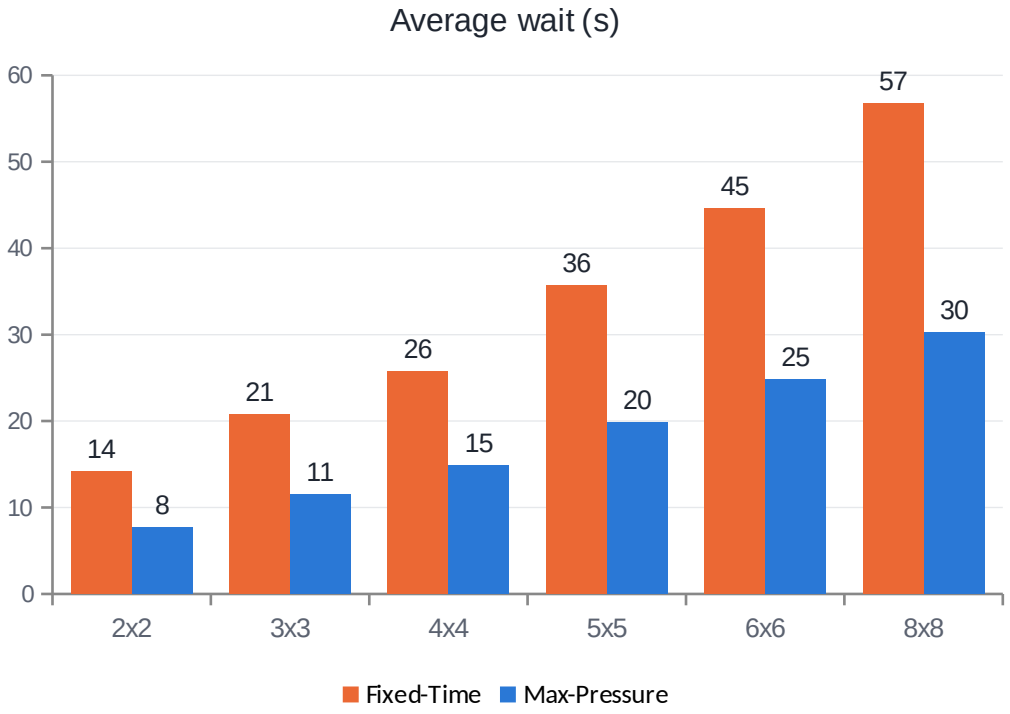
- src/agents: 6 agent classes
- src/algorithms: A*, Max-Pressure, hill climbing
- src/environment: roads, grids, simulator
- src/protocols: FIPA-ACL broker
- tests: 23 unit, 9 integration, 9 system

Scalability: 2x2 to 8x8

M1

Environment

S8: demand grows with grid size; every vehicle runs to its destination



461

agents on the 8x8 grid (64 junctions)

1.44 ms

per simulated second at 8x8: grows linearly

-42..47%

wait reduction at every size

Testing

M5

Coordinator

Review 2: demo quality & testing scenarios

43/43

automated tests pass

Level	Checks	#
Unit	A* admissible/consistent/optimal, Max-Pressure rules, BPR, broker	23
Integration	Protocols; safety: never green to green, yellow >= 3 s	9
System	Each scenario's acceptance criterion	9

11 defects found & fixed

Reason() ran twice per tick: timers at 2x

Departure times ignored

Ambulance baseline used a different controller

Preemption skipped yellow (unsafe)

Green-wave messages ignored

Vehicles omniscient about incidents

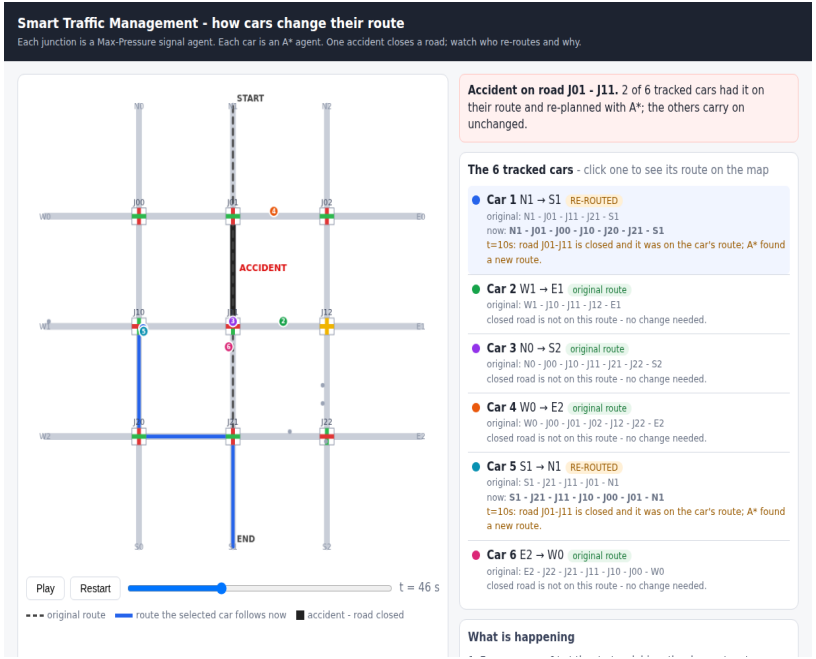
... each has a regression test

Live demo

What we will run in front of you

M1

Environment



1. **`run_demo.py`**
   Opens the page: 6 cars, 1 accident, who re-routes
2. **`--protocol`**
   Message log (ambulance, bus, incident)
3. **`--scenario 1..8`**
   Result tables for every experiment
4. **`--astar-trace`**
   A* expansion table
5. **`tests/run_tests.py`**
   43 tests in about 25 s

Limitations and next steps

TEAM

Whole team

What we did not model, and what we would add

Limitations (honest)

Two-phase signals, through movements only

Single-lane roads

Synthetic grid demand, not calibrated

Messages never delayed or lost

Green-wave helps corridors only

Bus priority doubles cross-street car wait

Ideas to extend

Q-learning junctions vs Max-Pressure

Contract-Net bidding between ambulances

Pedestrian agents and safe min-green

OpenStreetMap import (Coimbatore)

Message loss / delay robustness

Simulated annealing vs hill climbing



Decentralised agents, measurable gains

-47.5%
average wait

-29.6%
ambulance time

100%
optimal A* routes

43/43
tests pass

Every number is regenerated by python generate_results.py and checked by an automated test.

Questions?